

Sunidhi Pandit

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EDUCATION

Duke University (CGPA: 3.9/4.0)

Master of Engineering in Biomedical Engineering, Medical Device Design and Manufacturing Engineering

May 2026

Durham, NC

EXPERIENCE

Johnson & Johnson Medtech (Electrophysiology)

May 2025- January 2026

NPD Quality Engineer

Irvine, CA

- Engineered and validated an automated robotic test actuator interfaced with the TruePulse PFA generator, executing 100+ ablation cycles per catheter cable to eliminate manual testing, reducing validation cost by \$38K
- Led attribute-based DVT testing on 30 catheters (270 trials) using Minitab attribute agreement analysis, achieving 100% accuracy and demonstrating method repeatability/reproducibility for final commercial release.
- Authored a new standard test method (STM) for ablation performance testing, increasing measurement precision by 11% and repeatability by 21%, enabling consistent data generation and improved reliability across commercial catheter builds.
- Executed DVT for OmnyPulse PFA catheter integration with the CARTO mapping system, evaluating 4 critical performance parameters across 30 catheters to verify hardware-software interoperability before commercial release.
- Investigated a shaft-break non-conformance using Six Sigma DMAIC, identifying off-spec alloy composition as the source of mechanical failure, and avoiding ~\$42k in material scrap and rework costs.
- Maintained 10 risk management artifacts, including DFMEA and PFMEA, closing 14 high-severity risks by converting mitigations into testable design requirements with verified evidence.
- Authored an engineering report on a catheter lot deviation labelled “use-as-is” by performing worst-case risk assessment and tolerance analyses, leading to regulatory acceptance and a specification revision, saving ~\$15,000.
- Optimized polyurethane (PU) application process by designing controlled experiments to quantify material usage, and updated SOPs to reduce process variability by 17% and enhance manufacturing consistency.
- Detected process drift in catheter dimensions and used Minitab control charting, normality testing, and Cp/Cpk analysis to assess variability, driving targeted process changes that improved capability by 0.2 and reduced variation by 27%.
- Authored and deployed an FDA-compliant true-copy conversion process for electronically signed documents after an audit observation, establishing end-to-end traceability and strengthening inspection readiness.
- Designed arcing detection equipment to visualize minute arcing events, capturing previously undetectable electrical discharge events across 100 test runs and enabling root cause identification and design changes.

Duke Stimulation Centre

October 2024 - April 2025

Clinical R&D Engineer

Durham, NC

- Conducted over 70 hours of clinical observation and patient interviews to identify key usability and comfort challenges in existing CPAP therapy, translating unmet clinical needs into actionable design inputs for a next-generation tongue stabilization device.
- Applied structured design control and risk management (DFMEA, design reviews) to evaluate design safety, biocompatibility, and manufacturability, ensuring traceable alignment with FDA 510(k) pathways, ISO 13485 quality systems, and ISO 14971 risk standards.
- Designed, fabricated, and validated a functional prototype using silicone and carbon-fiber composites, and testing demonstrated a 20% improvement in patient comfort and consistent airway patency under simulated clinical conditions.
- Logged and categorized 24 use errors across formative evaluations, prioritizing 9 high-severity events and implementing targeted design controls that mitigated repeat occurrence before validation.

S&M Industries

May 2022 - July 2024

Senior Manufacturing Engineer

India

- Spearheaded ISO-controlled Class I-II cleanroom manufacturing across 2 production lines and 4 product variants (~10,000 units/month), sustaining >97% FPY and 98% OTIF through in-process quality controls.
- Led CAPA investigations for recurring manufacturing defects, coordinating cross-functional teams across quality, operations, and supply chain to identify root causes and implement corrective actions that reduced defect rate 35% and scrap 25%.
- Managed coordination with 3 global suppliers and cross-functional internal teams across quality, operations, and supply chain, driving manufacturing continuity and achieving \$40K in annual cost savings through supplier price negotiations.
- Implemented SPC-based process monitoring and capability tracking for device assembly operations, deploying fixtures and statistical controls that reduced assembly cycle time by 18% and increased equipment uptime by 12%.
- Performed IQ/OQ/PQ for new manufacturing equipment, verifying installation, establishing operating ranges, and confirming process capability for consistent production output.
- Directed Value Stream Analysis and Zero-Loss initiatives across manufacturing operations, identifying improvement projects that generated ~\$100k in cost savings over 2 fiscal years.
- Implemented SPC monitoring and statistical analysis (Minitab) across hybrid production lines, reducing chip-out scrap by 30% and improving process stability across multiple manufacturing operations.

PROJECT

Sleep-Apnea

September 2024 - April 2025

- Conducted over 70 hours of clinical observation and patient interviews to identify key usability and comfort challenges in existing CPAP therapy, translating unmet clinical needs into actionable design inputs for a next-generation tongue stabilization device.
- Designed and fabricated a functional prototype under \$1000 and conducted usability assessment testing, demonstrating an improvement in patient comfort and consistent airway patency under simulated use conditions.
- Conducted DFMEA and PFMEA risk assessments across 10+ potential failure modes, identifying critical design and process risks and implementing mitigation strategies to improve device safety and manufacturability.
- Developed and maintained the Design History File (DHF) for a novel sleep apnea therapy device, ensuring traceability across user needs, design inputs, verification testing, and risk management documentation.